Maestro: Manage. Master. Mitigate

SRCC Business Conclave 2026

Project SilicoSphere

Team: The Case Flayers

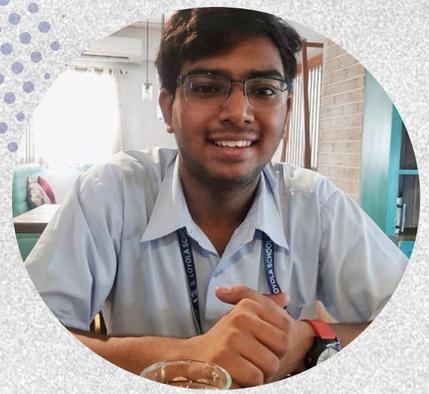
Balancing geopolitical risk mitigation with margin sustainability



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SilicoSphere Market Dynamics: Risk Exposure & Growth Opportunity

Supply chain concentration, sovereign demand, and resilience-driven growth

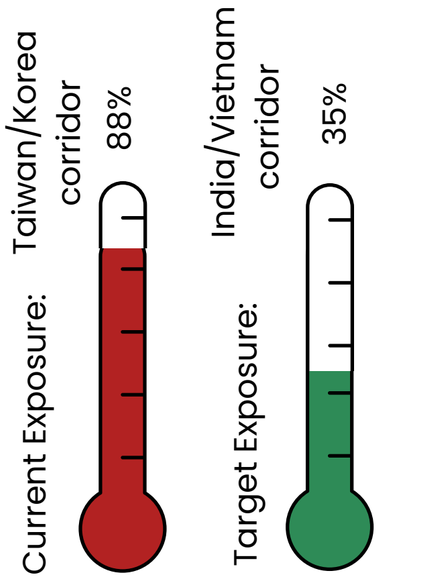
Key Exposure Facts

88% — Fabrication concentrated Taiwan–Korea corridor

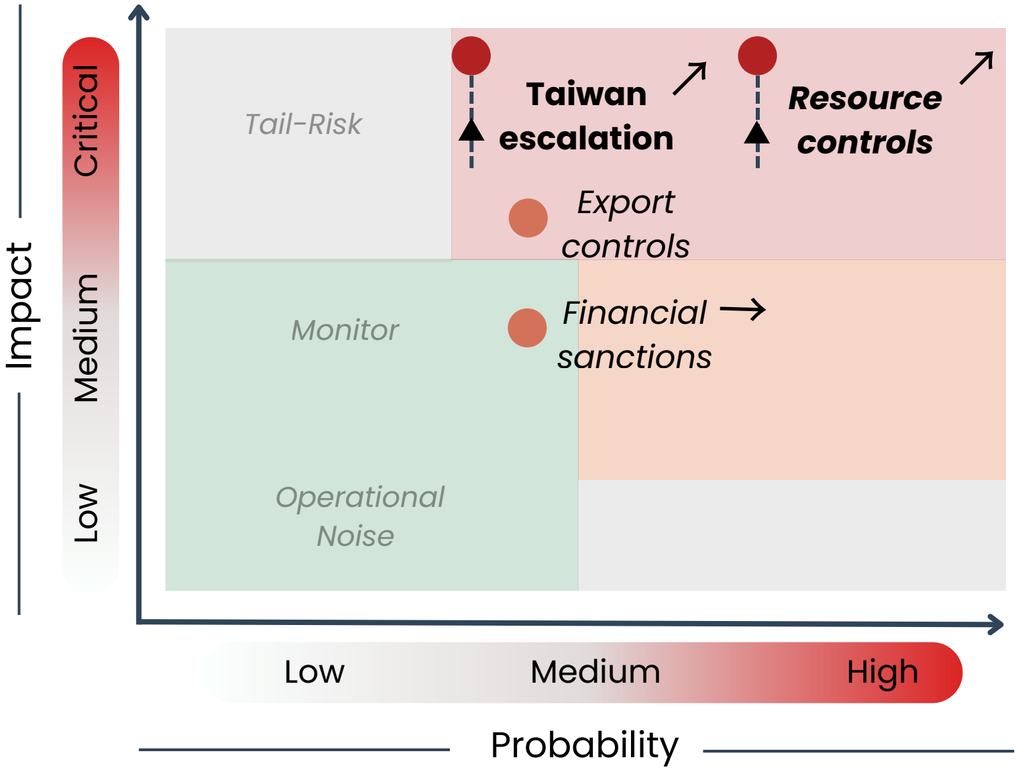
40% — Lead-time disruption risk Peak season exposure

$120M+ — Revenue at risk Single-region dependency

Strategic Pivot



Geopolitical Risk Priortisation



Risk exposure is clustering toward catastrophic zones

The Cost of Inaction

- **Revenue Wipeout:** $120M+ exposed to a single-point-of-failure.
- **Market Share Erosion:** 40% lead-time delay leads to "Seasonal Window Misses" and permanent brand switching.
- **Resource Paralysis:** 90% mineral dependency invites catastrophic upstream supply shocks.
- **Equity Discount:** Analysts apply a "Geopolitical Discount" to our P/E multiple due to concentration.

Strategic Mandate

Pivot to 'Risk-Adjusted Optionality' via a 70/30 Hybrid Model to prevent permanent brand erosion and market share loss.

Risk Analysis | **Sourcing Strategy** | **Product & Technology** | **Financials** | **Execution**

The 70/30 Dual-Node Architecture

From single-hub fragility to controlled continuity

Dual-Node Production Architecture

① 70% Strategic Continuity →

1. Taiwan–Korea core retained
2. 98% yield efficiency preserved
3. No transition shock

Result: Efficiency retained, geographic risk diluted.

② 30% Lifeboat Capacity →

1. Vietnam: ATP diversification
2. India: 28nm+ mature-node fab
3. 50% CAPEX support → 2026 rollout

Leveraging ISM fiscal support to offset CAPEX and protect shareholder IRR.

Upstream Resilience: The Mineral Hedge

Securing 6-month Gallium/Germanium stockpiles via Australian and Namibian corridors.

Feature	The Efficiency Hub (70%)	The Resilience Hub (30%)
Location	Taiwan / South Korea	India (FAB) + Vietnam (ATMP)
Node	Advanced (<7nm)	Mature (28nm+)
Goal	Profit & Performance	Continuity & Defense
Agility	Baseline	60-Day "Warm-Standby" Ramp

Agility Metric: The 60-Day Pivot

→ Trigger: Regional disruption detected.

→ Activation: Pre-qualified India/Vietnam "Warm-Standby" lines engaged.

→ Ramp: Full 30% capacity online within 8 weeks.

Outcome:

Efficiency retained while geographic risk is structurally diluted.

The Sovereign Tech Pivot: Scaling Margins & Autonomy

Decoupling from proprietary IP and volatile consumer markets to anchor long-term terminal value.

R&D Independence

IP Sovereignty: The RISC-V Shift

The Move: Phased migration from ARM/x86 to RISC-V (Open Source) architecture. yalty burdens.

The Win: Creates a "Sanction-Immune" design that can be manufactured on any compliant fab without ro

Market Pivot

The "Sovereign-Safe" Portfolio

The Move: Shift focus from low-margin consumer electronics to Strategic Infrastructure (Defense, 5G, Smart Grids).

The Win: High entry barriers via Trusted Foundry certifications and ITSAR standards.

Financial Hedge

Margin Stabilization Strategy

Segment	Volume	Margin	Revenue Stability
Consumer Chips	High	12–15%	Cyclical
Defense/ Govt Chips	Moderate	25–35%	5–10 Year Contracts

Defense margins anchor cash flows to fund geographic de-risking.

Ecosystem Leverage

Policy & Strategic Alliances

The Move: Deep integration with BEL and DRDO.

The Win: Utilizing DLI 2.0 policy to secure 50% reimbursement on all design R&D expenditure.

Net Strategic Transformation

- 🛡️ IP Sovereignty Secured
- 💰 High-Margin Revenue Anchored
- 📉 R&D Cost De-Risked
- 🤝 Public-Private Integration

Key Takeways

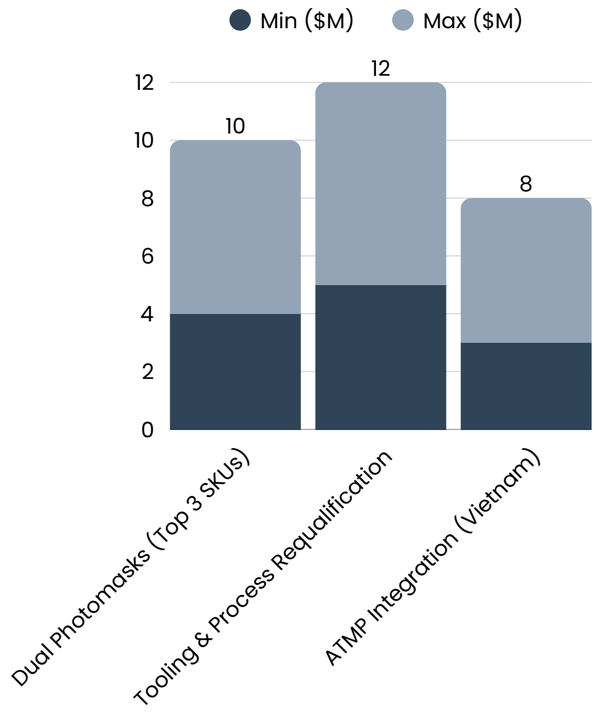
- IP Sovereignty: RISC-V secures designs against sanctions and royalty burdens.
- Market Pivot: Strategic infra shift captures stable 25–35% margins.
- Cash Flow Anchor: Long-term contracts fund and de-risk supply chain diversification.

We are moving from a Component Vendor to a Sovereign Tech Partner.

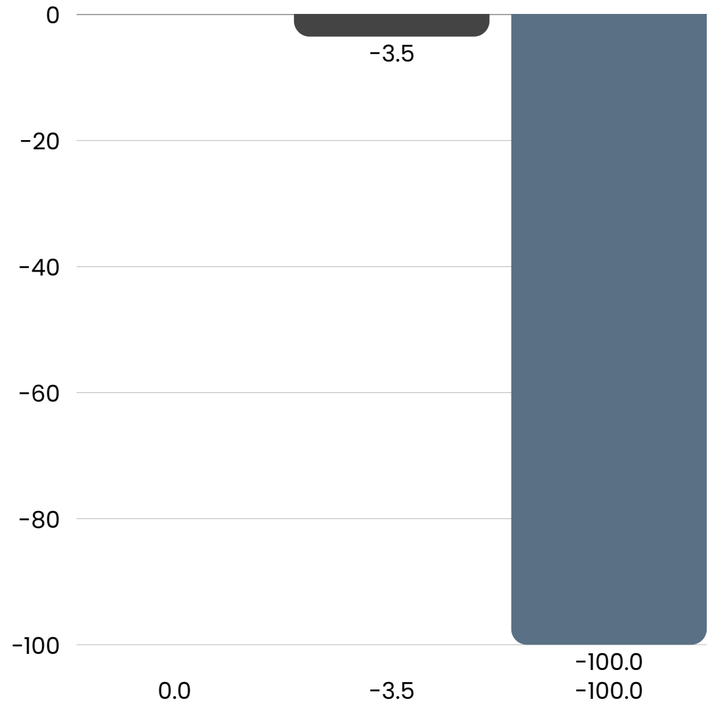
Strategic Capital Allocation: Financing Global Resilience

Utilizing sovereign subsidies to fund terminal value preservation and margin stabilization.

3-Year Incremental CAPEX



Yield Assumption: The "Resilience Premium"



This chart details the $15M–$25M capital required for dual-masking, requalification, and Vietnam ATMP setup. Darker shaded regions highlight our net exposure after 50% ISM subsidies, proving that sovereign incentives effectively halve our direct investment while doubling our strategic footprint.

The "Resilience Premium" Trade-off This visual frames our strategy as insurance: we trade a 3.5% margin dip (blue bar) to eliminate a 100% revenue wipeout risk (purple bar). This "Resilience Premium" secures $120M in annual revenue against geopolitical blockades, stabilizing terminal valuation.

Strategic Trade-off: 3.5% Margin Reduction vs. 100% Revenue Wipeout. This CAPEX is not cost inflation; it is Terminal Value Preservation funded by sovereign incentives.

SilicoSphere Execution Framework: Phased Resilience & Sovereign Scaling

Three-year transition from concentration risk to institutionalized agility

Phase 1: Foundation

Core Actions

- Secure India ATMP partnership
- Sign rare-earth sourcing MoUs
- Mirror masks for top 3 SKUs
- Hire Chief Resilience Officer
- Begin Trusted Foundry certification

KPI

- 15% diversified capacity secured
- Mineral exposure <60% single-source

Outcome

Minimum viable redundancy established

Phase 2: Qualification

Core Actions

- Tape-out first 28nm chip in India
- Achieve 85% yield ramp
- Operationalize 6-month mineral buffer
- Launch first sovereign-safe SKU
- Activate AI Supply Chain Control Tower

KPI

- <90-day ramp elasticity
- ISM subsidy drawdown initiated

Outcome

Functional dual-hub production model

Phase 3: Scale & Optimization

Core Actions

- Stabilize 70/30 hybrid split
- Expand defense revenue share to 20%
- Implement supplier concentration cap (<40%)
- Begin phased RISC-V architecture integration

KPI

- Geopolitical exposure score: 88% → 35%
- Gross margin stabilized within −3.5% band

Outcome

Resilient, sovereign-aligned semiconductor platform

Centralized Control Tower

AI-driven orchestration of:

- Multi-country production
- Logistics flows
- Yield tracking

Leadership Structure

- Chief Resilience Officer (reports to CEO)
- Dedicated cross-border SCM team
- Policy liaison for ISM & defense contracts

Risk Management

- Quarterly geopolitical stress test
- Scenario-based volume shift simulations
- Yield variance monitoring
- Raw material exposure dashboard

Annexure

- India Semiconductor Mission - https://ism.gov.in/
- Ministry of Electronics & IT (MeitY) semiconductor scheme announcements - https://www.meity.gov.in/
- Government of India official press releases - https://pib.gov.in/
- U.S. Department of Commerce – Bureau of Industry and Security (BIS) - https://www.bis.doc.gov/
- The White House CHIPS and Science Act materials - https://www.whitehouse.gov/

The resource nationalism references are based on: Ministry of Commerce of the People's Republic of China announcements on gallium/germanium export controls
Reporting from:

- Reuters
- Financial Times

- Taiwan Semiconductor Manufacturing Company (earnings calls & investor presentations) - https://www.tsmc.com/
- Semiconductor Industry Association - https://www.semiconductors.org/
- RISC-V International - https://riscv.org/
- Public semiconductor industry commentary on open ISA adoption.


THANK
YOU